SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 3

Experiments

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

FACULTY : Dr. D. SUDHAGAR

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 50

Code of Conduct:

I **G. Naveen Kumar Reddy** _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Naveen Kumar Reddy

QUESTIONS:10 **Total Marks:50**

1. Learning Algorithms

Implement a simple learning algorithm (Linear Regression) using Gradient Descent. Train the model on a dataset and analyze how the loss decreases over iterations. Plot the learning curve.

Ans:

- Initialize weights (w) and bias (b)
- Use cost function: Mean Squared Error (MSE)
- Update rule:

$$w = w - \alpha \frac{\partial J}{\partial w}$$

- Train on dataset and plot loss vs iterations

Observation:
Loss decreases gradually → shows convergence

2. Maximum Likelihood Estimation (MLE)

Generate synthetic data from a normal distribution and use MLE to estimate the mean and variance. Compare estimated values with actual parameters.

Ans:

- Generate data: $X \sim N(\mu, \sigma^2)$
- MLE formulas:

$$\hat{\mu} = \frac{1}{n}\sum x_i \,, \; \hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \hat{\mu})^2$$

Result:
Estimated mean and variance ≈ actual values (small error)

3. Building Machine Learning Algorithms

Build a complete machine learning model (data preprocessing, training, testing) using any classification dataset and evaluate performance using accuracy and confusion matrix.

Ans:

Steps:

1. Data preprocessing (handle missing values, normalization)
2. Train-test split
3. Train classifier (e.g., Logistic Regression)
4. Evaluate using:
   - Accuracy
   - Confusion Matrix

Conclusion:
Model performance depends on data quality and preprocessing

4. Neural Networks

Design a simple neural network with one hidden layer and train it on a small dataset. Analyze how the network learns non-linear patterns.

Ans:

- Architecture:
  - Input layer → Hidden layer → Output layer
- Use activation (ReLU / Sigmoid)

Observation:
Model learns non-linear patterns (unlike linear regression)

5. Artificial Neurons

Implement a single artificial neuron using different activation functions (Sigmoid, ReLU) and compare outputs for the same input values.

Ans:

- Output:

$$y = f(wx + b)$$

Compare:

- **Sigmoid:** Smooth, outputs (0,1)
- **ReLU:** Faster, outputs max(0, x)

Conclusion:
ReLU performs better in deep networks

6. Perceptron and Multilayer Perceptron (MLP)

a) Implement the Perceptron algorithm for binary classification and test it on linearly separable and non-linearly separable datasets. Analyze the results.

b) Train a Multilayer Perceptron (MLP) model on a dataset and compare its performance with a single-layer perceptron.

Ans:

(a) Perceptron
- Works for **linearly separable data**
- Fails for **non-linear data**

(b) MLP
- Multiple layers + activation functions
- Handles complex patterns

Conclusion:
MLP > Perceptron in performance

7. Forward Propagation and Backpropagation Algorithm

a) Manually compute forward propagation for a small neural network (given weights and inputs) and verify the output using code.

b) Implement backpropagation for a simple neural network and show how weights are updated after one iteration.

Ans:

(a) Forward Propagation
- Input → multiply weights → apply activation → output

(b) Backpropagation

- Compute error
- Update weights using gradient descent

Key Idea:
Weights updated to minimize loss

8. Gradient Descent and Stochastic Gradient Descent (SGD)

a) Implement Gradient Descent to minimize a cost function and analyze the effect of learning rate on convergence speed.

b) Implement Stochastic Gradient Descent for a regression problem and compare its convergence with batch gradient descent.

Ans:

(a) Gradient Descent

- Uses entire dataset
- Stable but slow

(b) SGD

- Uses single sample
- Faster but noisy

Conclusion:
SGD converges faster but fluctuates

9. Mini-Batch Gradient Descent

Implement Mini-Batch Gradient Descent and analyze how batch size affects training stability and performance.

Ans:

- Uses small batches
- Balance between GD and SGD

Observation:

- Small batch → noisy updates
- Large batch → stable but slow

Best practice: Medium batch size (e.g., 32, 64)

10. Curse of Dimensionality

Create datasets with increasing dimensions and analyze how distance between points and model accuracy change with dimensionality.

Ans:

- Increase dimensions → distance between points increases
- Data becomes sparse

Effects:

- Model accuracy decreases
- Computation increases

Conclusion:
High-dimensional data requires dimensionality reduction

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understandin g of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretatio n of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solv ing Approach	20	Logical, well-structure d, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized , and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand